From edge to bulk in free fermions in disguise

Explicit density correlators from induced paths

Istvan Vona

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Abstract

A free-fermion spectrum does not by itself determine local spin correlation functions. For the open free-fermions-in-disguise chain, however, each of the first three Hamiltonian terms, and by reflection each of the last three, has an explicitly computable infinite-temperature dynamical correlator with every term in the chain. We derive these correlators in three steps: the end-to-end density correlator, boundary sources coupled to a clique of up to three terms, and a target moved into the bulk. The mechanism is trace orthogonality: two induced-path products survive precisely when their vertex sets differ by the target vertex. At the endpoint this gives one scalar path polynomial; in the bulk only two terminal configurations and three residual interval polynomials are needed. Using unnormalized spectral residues removes all fermion-normalization factors, while a uniform subtraction accounts for stationary source components without casework. The resulting finite double sums require polynomial arithmetic work in system size. The derivation does not reconstruct a generic bulk operator or solve the bulk–bulk inverse problem.

1. The question and the chain

Fendley's open chain has a free-fermion spectrum even though its local terms are not ordinary Jordan–Wigner fermion bilinears [1, 2, 3]. The inverse problem asks for the converse dictionary: which local spin operators can be expressed simply using the hidden modes? Boundary-operator dynamics was obtained by Vona, Mestyán and Pozsgay [4]. The path-product expansion of Fukai, Pozsgay and Vona [5] now makes the operator content of the modes explicit. Here we use that expansion to evaluate density correlators without solving the full inverse problem for the bulk density.

Let

$$\mathcal{H} = \sum_{j=1}^{M} h_j, \qquad h_j = b_j g_j, \qquad B_j = b_j^2 > 0, \qquad g_j^2 = 1, \quad g_i g_j = (-1)^{\mathbf{1}_{0<|i-j|\le 2}} g_j g_i. \tag{1}$$

The couplings b_j may have either sign. The frustration graph has one vertex for each h_j and an edge for every anticommuting pair: it is the square of an open path, not the ordinary path. We fix the spin realization

$$g_j = Z_{j-2} Z_{j-1} X_j, \qquad Z_0 = Z_{-1} = 1, \qquad \langle A \rangle_\infty = 2^{-M} \operatorname{Tr} A, \qquad A(t) = e^{\mathrm{i}\mathcal{H}t} A e^{-\mathrm{i}\mathcal{H}t}. \tag{2}$$

The ordered words $g_A = \prod_{j\in A}^{\uparrow} g_j$ are trace-orthonormal: $\langle g_A^\dagger g_C \rangle_\infty = \delta_{A,C}$. Thus a nonempty word has zero trace. This is the trace convention used throughout; no projection onto a particular symmetry sector is made. Our observables are the *coupled terms* h_j. Divide $\langle h_i(t) h_j \rangle_\infty$ by $b_i b_j$ to obtain the correlator of g_i, g_j.

For an interval, define its weighted independence polynomial by

$$P_{[a,b]}(x) = \sum_{\substack{I \subseteq [a,b] \\ |i-j|>2 \ (i\neq j)}} (-x)^{|I|} \prod_{i\in I} B_i, \qquad P_{[a,b]} = 1 \quad (a > b). \tag{3}$$

Selecting or omitting the last vertex gives

$$P_{[a,m]}(x) = P_{[a,m-1]}(x) - x B_m P_{[a,m-3]}(x). \tag{4}$$

Write $P(x) = P_{[1,M]}(x)$ and $S = \lceil M/3 \rceil$. We first assume simple roots $x_a = u_a^2 > 0$, and use signed labels

$$\mathcal{I} = \{-S, \dots, -1, 1, \dots, S\}, \qquad u_{-a} = -u_a, \quad \varepsilon_s = u_s^{-1}, \quad P'_s = P'(u_s^2). \tag{5}$$

These are the single-particle energies: the spectrum is built from signed sums of the positive ε_a [1, 2]. Singular parameter choices are obtained by continuity of the final finite-dimensional correlator, not by substituting into an individual singular residue.

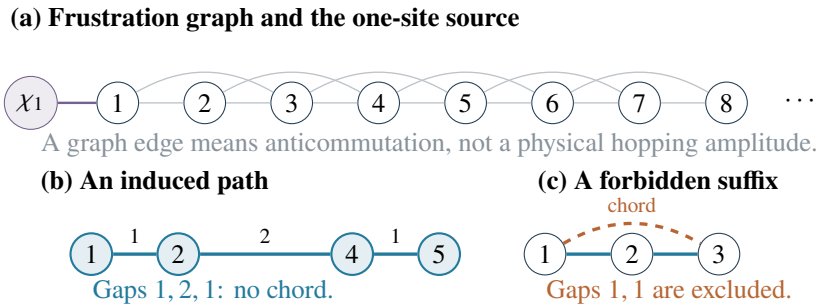


Figure 1. The graph behind the calculation. An induced path contains no edge between nonconsecutive path vertices. From a left boundary source, its physical vertices increase monotonically, with gaps 1 or 2, and never two successive gaps 1. These local rules will replace an exponentially large operator expansion by a scalar recurrence.

2. A boundary source turns dynamics into residues

Start with $\chi_1 = Z_1$. It anticommutes only with h_1, so

$$\mathcal{L}(A) := \tfrac{1}{2}[\mathcal{H}, A], \qquad \mathcal{L}(\chi_1) = h_1\chi_1, \qquad h_1 = \mathcal{L}(\chi_1)\chi_1. \tag{6}$$

Thus it suffices to evolve the source and its commutator. Add a source vertex α_1 adjacent only to vertex 1. A rooted induced path p has physical vertices $(i_1, \dots, i_n)$, and path word

$$\Pi_p = \chi_1 h_{i_1} \cdots h_{i_n}, \qquad \Pi_\varnothing = \chi_1. \tag{7}$$

Remove the path and every vertex adjacent to it, including those adjacent to the source. The remaining physical graph is the *residual graph* $\overline{p}$. For the empty path it is $[2, M]$; for a nonempty path ending at r it is $[r+3, M]$. The latter fact follows directly from the gap rule in Fig. 1.

The path-product numerator and its resolvent are

$$\Xi_1(u) = \sum_p (-u)^{|p|} P_{\overline{p}}(u^2)\Pi_p, \qquad \Phi_1(u) = \frac{\Xi_1(u)}{P(u^2)} = (1 - u\mathcal{L})^{-1}\chi_1. \tag{8}$$

Here $|p|$ counts physical vertices. The equality is the path-product identity of Ref. [5], specialized to this chain. Its useful local form is

$$u\mathcal{L}\Xi_1(u) = \Xi_1(u) - P(u^2)\chi_1. \tag{9}$$

It can be understood without diagonalizing a spin Hamiltonian. In a commutator, an internal path vertex has two path neighbors and hence contributes nothing. The endpoint can be removed or extended; the residual-polynomial recurrence accounts for these two operations. Terms that insert a chord cancel between the corresponding alternative parent paths. The surviving empty-path term is exactly $-P(u^2)\chi_1$. Ref. [5] proves this cancellation and the associated fermionic algebra; the correlator derivations below require only (9) and the explicit path words.

A normalization-free choice. Instead of introducing normalized fermions and their source coefficients separately, take the spectral components of χ_1 directly:

$$F_s^{(1)} = -\frac{\Xi_1(u_s)}{2u_s^2 P_s'}, \qquad \mathcal{L}F_s^{(1)} = \varepsilon_s F_s^{(1)}. \tag{10}$$

The sign and factor follow from the residue at $u = u_s$: $\operatorname{Res}\Phi_1 = \Xi_1(u_s)/(2u_s P_s')$, whereas $\operatorname{Res}(1 - u/u_s)^{-1}F_s^{(1)} = -u_s F_s^{(1)}$. Partial fractions therefore give

$$\chi_1(t) = Z_1^{\mathrm{st}} + \sum_{s\in\mathcal{I}} e^{2it\varepsilon_s} F_s^{(1)}, \qquad [\mathcal{H}, Z_1^{\mathrm{st}}] = 0, \qquad Z_1^{\mathrm{st}} = \chi_1 - \sum_s F_s^{(1)}. \tag{11}$$

The superscript distinguishes the stationary operator Z_1^{st} from the Pauli matrix Z_1. No explicit construction of this stationary part will be needed.

For comparison with normalized modes, set $Q_1(x) = P_{[2,M]}(x)$, $N_s^{(1)} = 2\sqrt{-u_s^2 Q_1(u_s^2)P_s'}$ and $\Psi_s^{(1)} = \Xi_1(u_s)/N_s^{(1)}$. Then $F_s^{(1)} = d_s^{(1)}\Psi_s^{(1)}$ with $d_s^{(1)} = 2Q_1(u_s^2)/N_s^{(1)}$ and $\{\Psi_s^{(1)}, \Psi_r^{(1)}\} = \delta_{s+r,0}$ [5]. Equation (10) is simply this product with its normalization factors cancelled.

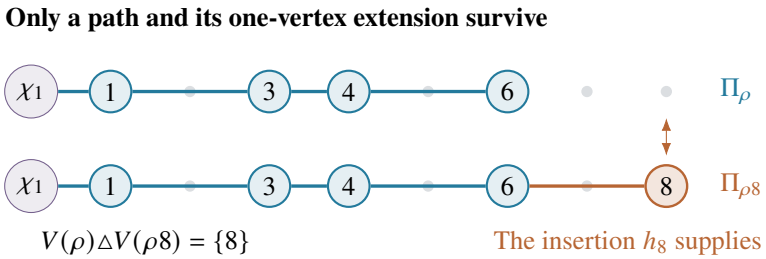


Figure 2. The end-to-end trace selection, illustrated at $M = 8$. The two paths have the same prefix $(1, 3, 4, 6)$; one ends at 6, and the other appends 8. The closed neighborhood of either path reaches the end of the chain. Reversing their order changes the trace sign, producing the antisymmetric factor $u - v$.

3. First calculation: the end-to-end correlator

3.1. One path polynomial

Let $R_m(y)$ sum all induced paths from 1 to m, weighting every physical vertex *except the final one* by $-yB_i$:

$$R_m(y) = \sum_{p:1\rightsquigarrow m} \prod_{i\in V(p)\setminus\{m\}} (-yB_i). \tag{12}$$

There are only two terminal possibilities: a final jump $m-2 \to m$, or a final pair of jumps $m-3 \to m-1 \to m$. Consequently

$$R_1 = 1, \qquad R_2 = R_3 = -yB_1, \qquad R_m = -yB_{m-2}R_{m-2} + y^2 B_{m-3}B_{m-1}R_{m-3} \quad (m \geq 4). \tag{13}$$

For example $R_4 = y^2 B_1(B_2 + B_3)$ and $R_5 = y^2 B_1 B_3 - y^3 B_1 B_2 B_4$. The terminal coupling B_m is absent: the observable insertion will supply it.

3.2. Trace orthogonality pairs a path with its extension

Consider the unnormalized overlap

$$\mathcal{K}_{1M}(u, v) := \langle \Xi_1(u)\Xi_1(v)h_M\rangle_\infty. \tag{14}$$

In a product of two path words the two source factors disappear. The trace can then be nonzero only if

$$V(p) \,\triangle\, V(q) = \{M\}, \tag{15}$$

where $\triangle$ is symmetric difference. Hence the paths are a common prefix ρ and its one-vertex extension ρM. Both residual graphs are empty: the prefix already ends at $M-1$ or $M-2$. This is why no right-tail polynomial remains.

To see the sign explicitly, let $n = |\rho|$ and $B_\rho = \prod_{i\in\rho} B_i$. The two traces are

$$\langle \Pi_\rho \Pi_{\rho M} h_M\rangle_\infty = (-1)^n B_M B_\rho, \qquad \langle \Pi_{\rho M}\Pi_\rho h_M\rangle_\infty = (-1)^{n-1} B_M B_\rho. \tag{16}$$

Multiplying by $(-u)^n(-v)^{n+1}$ and $(-u)^{n+1}(-v)^n$, respectively, gives $B_M(u-v)(-uv)^n B_\rho$. Summing the prefixes proves

$$\boxed{\mathcal{K}_{1M}(u, v) = B_M(u - v)R_M(uv).} \tag{17}$$

This identity holds for arbitrary u, v, not just at spectral roots.

3.3. A uniform treatment of stationary components

From (6), $h_1(t) = (\sum_s \varepsilon_s e^{2it\varepsilon_s} F_s^{(1)})\chi_1(t)$. The remaining stationary matrix element can be eliminated at time zero. For $M > 1$,

$$\langle F_s^{(1)}\chi_1 h_M\rangle_\infty = 0, \qquad \langle F_s^{(1)} Z_1^{\mathrm{st}} h_M\rangle_\infty = -\sum_r \langle F_s^{(1)} F_r^{(1)} h_M\rangle_\infty. \tag{18}$$

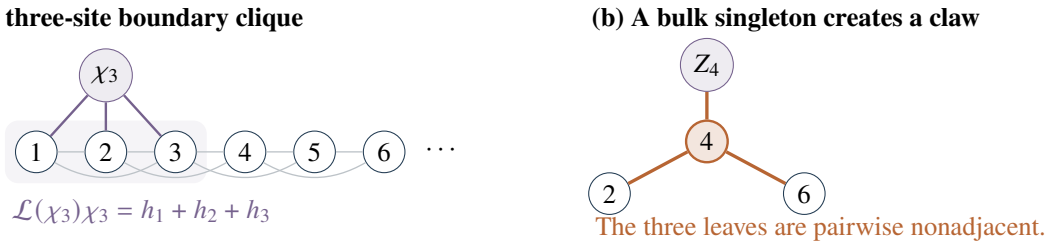


Figure 3. Why three boundary generators are accessible by this construction. The prefix cliques K_1, K_2, K_3 give valid sources, and differences of their density sums isolate h_1, h_2, h_3. By contrast, a source attached only to an interior vertex such as 4 creates the displayed induced claw for $M \geq 6$; the boundary-source theorem cannot be invoked there. The right-hand drawing shows only the induced four-vertex subgraph.

Indeed, a single source path cannot have physical vertex set $\{M\}$. The first equality is a selection rule, not an assumption that the stationary part vanishes. Define the common spectral weight

$$\mathcal{W}_{sr}(t) = \frac{e^{2it(\varepsilon_s+\varepsilon_r)} - e^{2it\varepsilon_s}}{4u_s^3 u_r^2 P_s' P_r'}. \tag{19}$$

Equations (10), (17) and (18) yield the first answer:

$$\boxed{\langle h_1(t) h_M \rangle_\infty = B_M \sum_{s,r\in\mathcal{I}} \mathcal{W}_{sr}(t)(u_s - u_r) R_M(u_s u_r), \qquad M > 1.} \tag{20}$$

When $Z_1^{\mathrm{st}} = 0$, the subtraction in (19) sums to zero; when it is present, the same subtraction includes its effect exactly. No congruence-class switch is needed. Moreover $\mathcal{W}_{sr}(0) = 0$, so the off-diagonal equal-time answer vanishes term by term. Pairing (s, r) with $(-s, -r)$ replaces the exponentials by cosines, showing that the result is real and even in t.

For $M = 2, 3$, the single positive root gives the elementary check

$$\langle h_1(t) h_M \rangle_\infty = \frac{B_1 B_M}{\sum_{i=1}^M B_i} \left[1 - \cos\left(2t\sqrt{\sum_{i=1}^M B_i}\right)\right]. \tag{21}$$

For $M = 1$, separately, $\langle h_1(t) h_1 \rangle_\infty = B_1$.

4. The next step: the first three boundary terms

A source for h_2 alone is not needed. Instead use the nested boundary sums

$$K_k = \{1, \ldots, k\}, \quad S_k = h_1 + \cdots + h_k, \quad \chi_k = Z_1 \cdots Z_k, \quad \mathcal{L}(\chi_k) = S_k \chi_k, \qquad 1 \leq k \leq \min(3, M). \tag{22}$$

Each K_k is a *simplicial clique*: its vertices are mutually adjacent, and for every vertex in K_k its neighbors outside K_k form a clique. Thus attaching α_k to K_k preserves the graph condition needed for the path-product construction [2, 3, 5]. A path may start at any vertex of K_k, but it cannot visit another one: that would create a chord back to the source. After leaving the clique, the same monotone gap rule applies.

Replace χ_1 and its paths by χ_k and paths rooted at α_k in (8). Denote the result $\Xi_k(u)$. The empty-path residual becomes $Q_k(x) = P_{[k+1,M]}(x)$; a nonempty path ending at r still has residual $[r + 3, M]$. Nothing changes in the spectrum or residue denominator:

$$F_s^{(k)} = -\frac{\Xi_k(u_s)}{2u_s^2 P_s'}, \quad \chi_k(t) = Z_k^{\mathrm{st}} + \sum_s e^{2it\varepsilon_s} F_s^{(k)}, \quad S_k(t) = \left(\sum_s \varepsilon_s e^{2it\varepsilon_s} F_s^{(k)}\right) \chi_k(t). \tag{23}$$

Different k generally give different mode operators, even though their energies agree. We compute each clique correlator in its own source basis and only then take differences.

Let $R_m^{(k)}$ be the source-to-m path sum, again excluding the source and the final vertex from the weight. Set $R_m^{(k)} = 1$ for $1 \le m \le k$ and $R_m^{(k)} = 0$ for $m \le 0$. For $m > k$, resolve the two final gaps:

$$\begin{aligned} \rho_{2,m}^{(k)}(y) &= -yB_{m-2}R_{m-2}^{(k)}(y), \\ \rho_{1,m}^{(k)}(y) &= \begin{cases} -yB_k, & m = k+1, \\ y^2B_{m-3}B_{m-1}R_{m-3}^{(k)}(y), & m \ge k+2, \end{cases} \\ R_m^{(k)}(y) &= \rho_{1,m}^{(k)}(y) + \rho_{2,m}^{(k)}(y). \end{aligned} \tag{24}$$

Here an out-of-range factor B_i with $i \le 0$ means zero. The special first line of the piecewise definition is the direct path $\alpha_k \to k \to k+1$. It must not be replaced by a fictitious preceding physical step. Farther into the chain, (24) is exactly the scalar recurrence (13). In particular

$$R_3^{(2)} = -y(B_1 + B_2), \qquad R_4^{(3)} = -y(B_2 + B_3). \tag{25}$$

The endpoint result immediately generalizes by replacing R_M with $R_M^{(k)}$ in (20), provided $M > k$.

Which source has a stationary part? It is the *source*, not just the chain length, that decides this. The resolvent identity implies [5]

$$(Z_k^{\mathrm{st}})^2 = \lim_{x\to\infty} \frac{Q_k(x)}{P(x)}\,\mathbf{1}. \tag{26}$$

Thus $Z_1^{\mathrm{st}} \neq 0$ for $M \equiv 0, 2 \pmod 3$; $Z_2^{\mathrm{st}} \neq 0$ only for $M \equiv 0 \pmod 3$; and $Z_3^{\mathrm{st}} = 0$. The stationary content therefore depends on which boundary source is chosen. The uniform weight (19) avoids that issue altogether.

5. Moving the target into the bulk

Let $j > k$. The same trace selection rule now reads $V(p) \triangle V(q) = \{j\}$. The target must again be the final vertex of one path. It cannot be removed from the interior: the vertices on either side would have to become adjacent, contradicting inducedness of the longer path. Nor can the first physical vertex be removed, since the next vertex lies outside the source clique. Hence the pair is still $(\rho, \rho j)$.

What changes is only the residual graph. Define

$$L_j(x) = P_{[j+3,M]}(x), \qquad C_{2,j}(x) = P_{[j+1,M]}(x), \qquad C_{1,j}(x) = P_{[j+2,M]}(x). \tag{27}$$

The longer path always leaves L_j. If the shorter path ends at $j-2$, its residual is $C_{2,j}$; if it ends at $j-1$, its residual is $C_{1,j}$. These are the only possibilities, depicted in Fig. 4.

Write

$$\Delta_{a,j}(u,v) = uL_j(u^2)C_{a,j}(v^2) - vC_{a,j}(u^2)L_j(v^2), \qquad a = 1, 2. \tag{28}$$

The path-pair overlap is then

$$\boxed{\mathcal{K}_{kj}(u,v) := \langle \Xi_k(u)\Xi_k(v)h_j\rangle_\infty = B_j\left[\rho_{2,j}^{(k)}(uv)\Delta_{2,j}(u,v) + \rho_{1,j}^{(k)}(uv)\Delta_{1,j}(u,v)\right], \quad j > k.} \tag{29}$$

To derive it, repeat the endpoint sign calculation for each prefix ρ. In one ordering the extra factor $-u$ belongs to the longer path, whose residual is $L_j(u^2)$; the shorter path carries $C_{a,j}(v^2)$. The reversed ordering gives the other term with the opposite sign. Summing the common prefix weights gives $\rho_{a,j}^{(k)}$. This proves (29) for arbitrary spectral parameters. It also explains why a single polynomial in uv no longer suffices: the two residual tails are evaluated separately at u^2 and v^2.

Theorem 1 (Boundary-clique–bulk correlator)**.** *For $1 \le k \le \min(3, M)$ and $j > k$, let $\mathcal{W}$ be (19), let the path weights be generated by (24), and let the tails be (27). Then*

$$\boxed{D_{kj}(t) := \langle S_k(t)h_j\rangle_\infty = \sum_{s,r\in\mathcal{I}} \mathcal{W}_{sr}(t)\mathcal{K}_{kj}(u_s, u_r).} \tag{30}$$

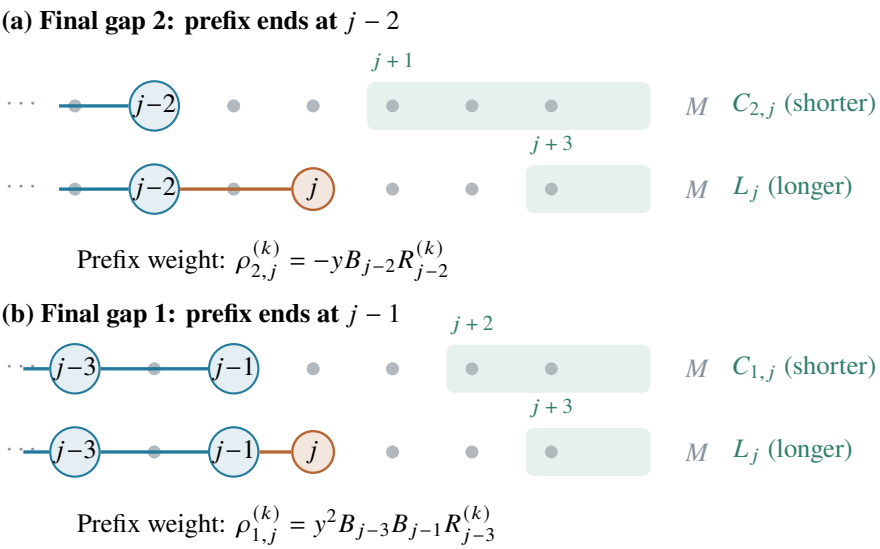


Figure 4. The entire bulk modification is local at the target. Shaded intervals are the residual graphs, not part of the selected path. Adding j shifts the start of the residual interval to $j+3$. The second prefix-weight formula is the generic one; at $j = k+1$ the prefix is just the clique vertex k, and its weight is $-yB_k$ as in (24). The labels $j \pm a$ denote positions along the chain; grey dots are unselected vertices.

Proof. Equation (23) reduces the calculation to two source components. Since $j \notin K_k$, no single source path has physical vertex set $\{j\}$, and therefore $\langle F^{(k)}_s \chi_k h_j\rangle_\infty = 0$. Eliminate Z^{st}_k exactly as in (18). Finally,

$$\langle F^{(k)}_s F^{(k)}_r h_j\rangle_\infty = \frac{\mathcal{K}_{kj}(u_s,u_r)}{4u_s^2u_r^2P'_sP'_r}.$$

Multiplication by ε_s and the time phases gives (30). □

Two useful checks. At $j = M$ all three tails are 1, so $\mathcal{K}_{kM} = B_M(u-v)R^{(k)}_M(uv)$, recovering the endpoint calculation. At the other extreme, $k = 3, j = 4$ gives $\rho^{(3)}_{2,4} = -yB_2$ and $\rho^{(3)}_{1,4} = -yB_3$. Thus the formula includes the first site beyond the boundary block; no artificial condition $j \geq k+3$ is required. The exceptional initial weight in (24) is essential for this extension.

6. Individual generators, reflection, and nearby targets

With $D_{0j} = 0$, taking differences of the nested source sums gives the desired densities:

$$\boxed{\langle h_m(t)h_j\rangle_\infty = D_{mj}(t) - D_{m-1,j}(t), \qquad m = 1,2,3, \quad j \geq 4.} \tag{31}$$

Both terms use the same roots of P, but their path weights are those of their own source cliques. This is the partial inverse problem in its most useful form: the dynamics of the boundary observable is simple, and only the appropriate two-source-component overlap of the bulk observable is needed. We never assert that h_j itself is quadratic in a fixed set of hidden modes.

For reflected couplings $\bar{b}_i = b_{M+1-i}$, write $\bar{D}$ for the same formulas evaluated on the reflected chain. Trace cyclicity gives

$$\langle h_j(t)h_{M-r+1}\rangle_\infty = \bar{D}_{r,M+1-j}(-t) - \bar{D}_{r-1,M+1-j}(-t), \qquad r = 1,2,3. \tag{32}$$

Thus the last three terms are equally accessible. In this real representation the correlators are even in time, so the minus sign may also be dropped after evaluation.

Targets inside the source clique. For completeness, the phrase "every term" includes $j \leq k$. In this case the only surviving pair is the empty path and the one-vertex path (j). Consequently the kernel is instead

$$\mathcal{K}_{kj}(u,v) = B_j\left[uL_j(u^2)Q_k(v^2) - vQ_k(u^2)L_j(v^2)\right], \qquad j \leq k. \tag{33}$$

Now $\langle \Xi_k(u)\chi_k h_j\rangle_\infty = uB_jL_j(u^2)$ is nonzero. Keeping this elementary term when eliminating the stationary part gives

$$D_{kj}(t) = \sum_{s,r} \mathcal{W}_{sr}(t)\mathcal{K}_{kj}(u_s,u_r) - \frac{B_j}{2}\sum_s \frac{L_j(u_s^2)}{u_s^2 P_s'} e^{2it/u_s}, \qquad j \le k. \tag{34}$$

Equations (30) and (34), followed by the same finite difference, therefore cover every target $1 \le j \le M$, including autocorrelations. At $t = 0$ they give $D_{kj}(0) = B_j \mathbf{1}_{j\le k}$.

Opposite boundary blocks need no new machinery. For $T_\ell = \{M-\ell+1,\dots,M\}$ and $M \ge 6$, summing (29) over the target block gives

$$\begin{aligned}
\langle \Xi_k(u)\Xi_k(v)\sum_{j\in T_\ell} h_j\rangle_\infty &= (u-v)\mathcal{R}_{k\ell}(uv),\\
\mathcal{R}_{k1}(y) &= B_M R_M^{(k)}(y),\\
\mathcal{R}_{k2}(y) &= B_{M-1}R_{M-1}^{(k)}(y) - yB_MB_{M-2}R_{M-2}^{(k)}(y),\\
\mathcal{R}_{k3}(y) &= B_{M-2}R_{M-2}^{(k)}(y) - yB_{M-1}B_{M-3}R_{M-3}^{(k)}(y).
\end{aligned} \tag{35}$$

These are first-hit path sums: the first vertex encountered in the target clique carries B_j, while each preceding vertex carries $-yB_i$. An induced path can first enter T_2 at $M-1$, or jump from $M-2$ to M; it can first enter T_3 at $M-2$, or jump from $M-3$ to $M-1$. These are exactly the displayed terms. In the overlap sum, contributions that continue inside the clique cancel the residual-tail terms. Equivalently, substitution of the degree-zero and degree-one endpoint tails into (29), followed by (24), verifies all three identities directly. The correlator is $\sum_{s,r}\mathcal{W}_{sr}(t)(u_s-u_r)\mathcal{R}_{k\ell}(u_su_r)$; differences in both block sizes isolate any of the nine opposite-edge pairs.

7. Computation, checks, and the extent of the result

Polynomial work, without enumerating paths. The path expansion is a proof device, not the computational algorithm. Build P by (4) and determine its $S = O(M)$ roots. At each root, generate all right tails by the backward recurrence

$$P_{[a,M]}(x) = P_{[a+1,M]}(x) - xB_aP_{[a+3,M]}(x). \tag{36}$$

For each signed pair (s,r), run the scalar recurrence (24) at $y = u_su_r$, evaluate the desired kernel, and sum (30). For a fixed edge–target pair this takes $O(MS^2) = O(M^3)$ arithmetic operations after the spectral data are obtained; storing its $O(S^2)$ coefficients reduces each additional time evaluation to $O(S^2)$. All targets can share the same path sweeps. A conventional polynomial-coefficient/root-finding implementation also has polynomial arithmetic cost. These are arithmetic bounds, not uniform bit-complexity or floating-point stability guarantees: large chains, strongly separated couplings, or clustered roots may require higher precision.

Independent verification. The path recurrence, resolvent identity, all-target overlap, and opposite-clique sums were checked by enumerating induced paths and multiplying graph-Clifford words in exact integer arithmetic, through $M = 18$ (102, 51, 509, and 117 checks, respectively). The finite-time formulas were separately compared with direct commutator evolution on the ordered-word basis for $2 \le M \le 12$, all allowed k, j, uniform, nonuniform and random signed couplings, and four times. Across 2,748 scalar comparisons the maximum absolute discrepancy was 1.05×10^{-13}. Reproducible scripts and parameter choices accompany the source files. These checks support, but do not replace, the analytic derivation. Useful implementation tests are

$$\langle h_ih_j\rangle_\infty = B_i\delta_{ij}, \qquad \sum_{j=1}^{M}\langle h_i(t)h_j\rangle_\infty = B_i, \tag{37}$$

the second following from conservation of $\mathcal{H}$.

Relation to existing methods. The spectral polynomial and hidden fermions originate in Refs. [1, 2, 3]; boundary inverse formulas and selected real-time correlators were obtained in Ref. [4]. The essential input here is the explicit path-product resolvent of Ref. [5], whose general proof is not repeated. The additional reduction is at the level of the *density overlap*: trace orthogonality leaves two terminal path configurations, and hence the scalar formulas (29)–(30).

Polynomial contraction is also consistent with the matrix-product-operator approach to mode expectation values developed for FFD circuits in Ref. [6]. In the Hamiltonian setting, the finite-range independent-set transfer recurrence has a finite-memory tensor-network representation, so a fixed product of mode operators with a local insertion can likewise be contracted with polynomial work. This gives an alternative complexity argument; the path calculation provides the explicit scalar kernel rather than a claim of first establishing polynomial simulability for boundary observables. Recent work on bulk generalized-Gibbs-ensemble expectations and quench dynamics [7] addresses a different question from the finite-time, infinite-temperature cross-correlators considered here. The distinction between the physical-mode algebra and degeneracy degrees of freedom is developed in Ref. [8].

What is not solved. The three-site reach comes from the nested boundary cliques, not from a claim that all densities have a quadratic expression in one common fermion basis. The block $\{1, 2, 3, 4\}$ is not a clique, so the same argument does not produce S_4. Attaching a source only to a bulk site generally creates a claw, as Fig. 3(b) shows. Thus, when both densities lie outside the accessible edge blocks, a correlator such as $\langle h_4(t) h_{M-3}\rangle_\infty$ for $M \geq 10$ is not determined by these formulas. This is a limitation of the construction, not an impossibility theorem for bulk–bulk dynamics. In particular, no truncation of a generic bulk operator to quadratic or quartic physical-fermion sectors has been assumed.

The central lesson is that a correlation function can be simpler than the inverse problem for either of its factors in a fixed fermionic dictionary. A boundary source supplies the dynamics; the normalized trace selects a small part of the target operator; and induced-path geometry evaluates that part with scalar recurrences. This is precisely the partial inverse problem solved here.

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